Database Management Systems
CSC_51053_EP

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Outline for today

0. Course goals and organization

1. Overview of databases and DBMSs

2. Data modeling

Course goals (1/2)

1. Gain **strong competencies** as a **user** of a relational database management system (RDBMS)

- Focus on Relational DBMSs as they are the most mature
- Designing a database
- Analyzing a database design, fixing its flaws
- SQL fluency: at the end of the course you should be capable to express
  - in a **machine-understandable language (SQL)**
  - a **complex request for information**
  - to be answered by a system holding a **very large database**
- SQL fluency ~ 25% of the overall effort

Course goals (2/2)

2. Teach the **main concepts and algorithms** behind the good properties of RDBMSs

- **scalability** to very large data size
- **strong concurrency control**

Understanding how RDBMSs work makes it easier to get excellent service from them.

Also, RDMBS algorithms are the foundation of more recent data management systems used by any large modern institution.

Databases today

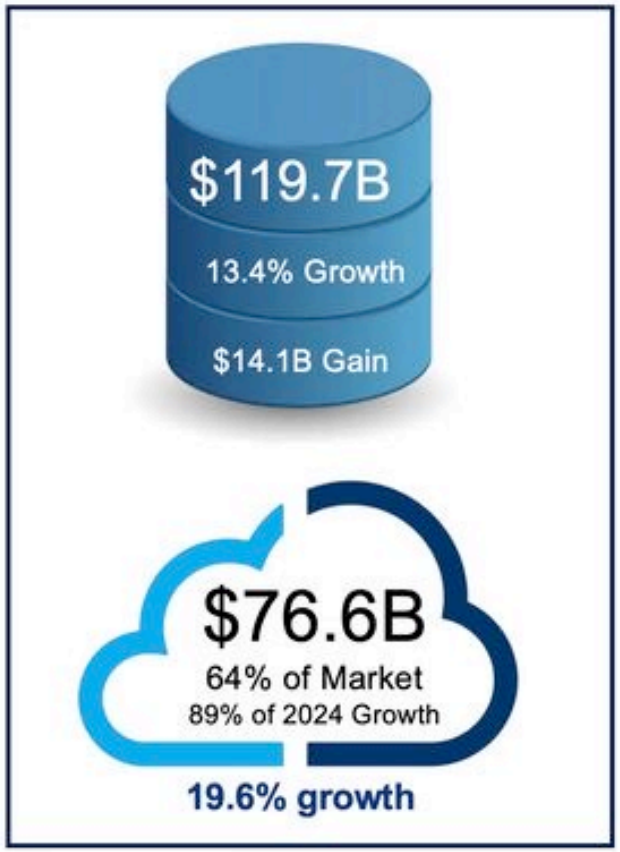
- This course corresponds to 1-2 courses in other institutions, e.g.
  - "Introduction to DB" (user perspective only)
    - Searching for information in the data
  - "Advanced Databases":
    - Query optimization: how is performance achieved?
    - Concurrency control and transactions: how to support multiple simultaneous users?

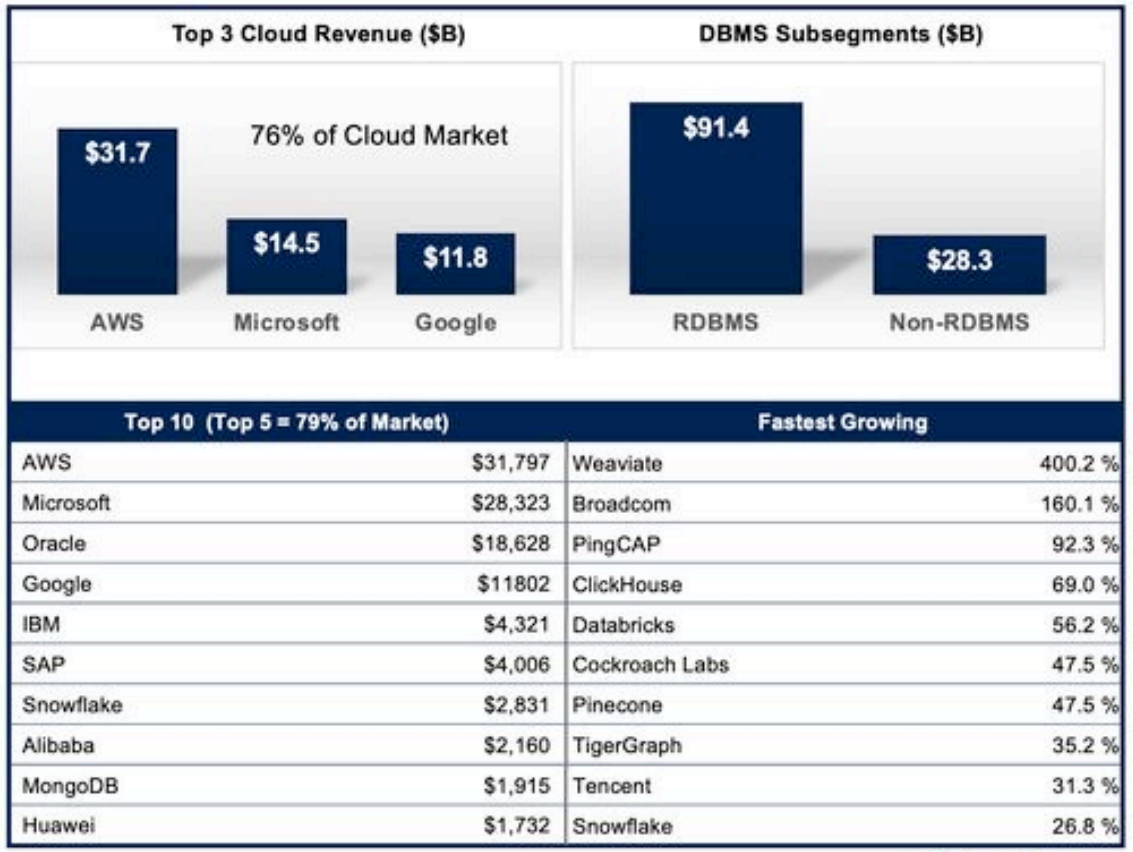
- At least one (and usually 2) are part of any CS engineering degree across the world

Databases today

- A prosperous industry, too

2024 DBMS Market Snapshot





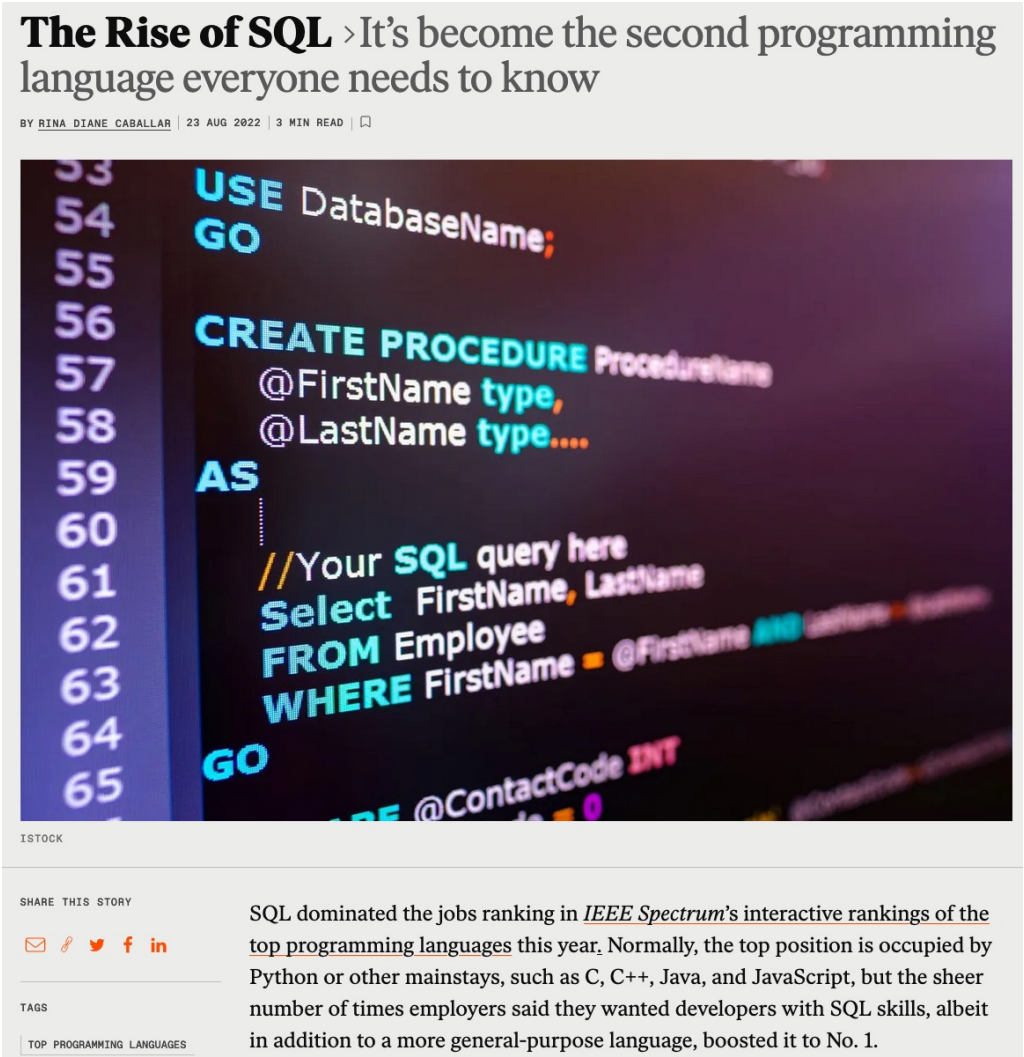
Top 10 (Top 5 = 79% of Market)		Fastest Growing	
AWS	$31,797	Weaviate	400.2 %
Microsoft	$28,323	Broadcom	160.1 %
Oracle	$18,628	PingCAP	92.3 %
Google	$11802	ClickHouse	69.0 %
IBM	$4,321	Databricks	56.2 %
SAP	$4,006	Cockroach Labs	47.5 %
Snowflake	$2,831	Pinecone	47.5 %
Alibaba	$2,160	TigerGraph	35.2 %
MongoDB	$1,915	Tencent	31.3 %
Huawei	$1,732	Snowflake	26.8 %

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Databases today

- This course corresponds to 1-2 courses in other institutions, e.g., “Introduction to DB” (user perspective only) + “Advanced Databases”
- At least one (and usually 2) are part of any CS engineering degree across the world
- A prosperous industry, too



The Rise of SQL ›It’s become the second programming language everyone needs to know

BY RINA DIANE CABALLAR | 23 AUG 2022 | 3 MIN READ

```
USE DatabaseName;
GO

CREATE PROCEDURE ProcedureName
    @FirstName type,
    @LastName type....
AS

    //Your SQL query here
    Select  FirstName, LastName
    FROM Employee
    WHERE FirstName = @FirstName
GO
```

ISTOCK

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TAGS

TOP PROGRAMMING LANGUAGES

SQL dominated the jobs ranking in *IEEE Spectrum*’s interactive rankings of the top programming languages this year. Normally, the top position is occupied by Python or other mainstays, such as C, C++, Java, and JavaScript, but the sheer number of times employers said they wanted developers with SQL skills, albeit in addition to a more general-purpose language, boosted it to No. 1.

https://spectrum.ieee.org/the-rise-of-sql

COURSE ORGANIZATION

Instructors and Contact Information

- **Ioana Manolescu** (Inria and Polytechnique)
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  - Office: 1105, Turing building (Inria/LIX)



- **Madhulika Mohanty** (Inria and Polytechnique)
  - Email: madhulika.mohanty@inria.fr
  - Office: 1028, Turing building (Inria/LIX)



Communications during the course

- Talk to us on Fridays
- Moodle mailing list and newsgroups
  - Preferred, so that a given question is answered once
- E-mail

Course organization

- Each week, 2 hours of **lecture**
- Slides in Moodle. Minor updates up to the same day.
    - Printed matter and own notes will be allowed at the exam.
    - No computers/phones will be allowed.
- Each week, 2 hours for:
    - **Exercises** on lecture material and/or
    - **Lab work** (on your laptop)
    - As per the school request, we will record exercise/lab attendance

Grading

Exam on Dec 12, 2025 (**9/20** points)

- ❑ *All paper material allowed then*
- ❑ *No electronics allowed*

Three short evaluations, each of 30 minutes, for **3/20** points each, during weekly exercise sessions

- ❑ *All paper material allowed then*
- ❑ *No electronics allowed*

Participation to PC (**2/20** points)

Solving and explaining exercises at the board

New grading (2025)

A+	Exceptional	The course's academic objectives were surpassed and the student has gone above and beyond expectations
A	Excellent	The course's academic objectives were fully attained and the student has demonstrated particular capabilities during the course
A-	Very good	The course's academic objectives were fully attained
B+	Good	The course's academic objectives were mostly attained
B	Fairly good	The course's academic objectives were generally attained with some strong performances
B-	Satisfactory	The course's academic objectives were generally attained with some areas needing improvement
C	Marginal pass	The course's academic objectives were partially attained with adequate knowledge to move on to more advanced work
D	Insufficient	The course's academic objectives were minimally attained with persistent weaknesses
F	Fail	The course's academic objectives were not attained or the student exhibited inappropriate behavior

Grade Point Scale is defined as follows

Literal grade	Grade Point Scale
A+	4,3
A	4,0
A-	3,7
B+	3,3
B	3,0
B-	2,7
C	2,0
D	1,0
F	0

New grading (2025)

- Grades from **C** to **A+** are passing grades that allow students to earn credits for the course.
- **D** is a failing grade. However, a D can be *compensated* for by a B or a higher grade (B- does not compensate), which, in the case of compensation, allows the student to earn credits for the course.

- **F** is a failing grade, non-compensable. Registration for a make-up exam is then mandatory when the course is eligible for a make-up exam.

The place of GenerativeAI

- Some AI technologies show genuine success; others still evolve
- The goal of this course (and the programs it is part of) is that the students acquire **new thinking patterns**, so that you can **competently address a data management problem** you may encounter in the future
  - We have always authorized all paper material during evaluations
  - We measure your ability to work with the taught material, not to reproduce it, or ”write SQL”

Learning changes your brain (for the better)

Neuroplasticity

33 languages

Article Talk Read Edit View history Tools

From Wikipedia, the free encyclopedia

"Neural plasticity" redirects here. For the journal, see Neural Plasticity (journal). For the 2014 Cold Specks album, see Neuroplasticity (album).

Neuroplasticity, also known as **neural plasticity** or just **plasticity**, is the medium of neural networks in the brain to change through growth and reorganization. Neuroplasticity refers to the brain's ability to reorganize and rewire its neural connections, enabling it to adapt and function in ways that differ from its prior state. This process can occur in response to learning new skills, experiencing environmental changes, recovering from injuries, or adapting to sensory or cognitive deficits. Such adaptability highlights the dynamic and ever-evolving nature of the brain, even into adulthood.[1] These changes range from individual neuron pathways making new connections, to systematic adjustments like cortical remapping or neural oscillation. Other forms of neuroplasticity include homologous area adaptation, cross modal reassignment, map expansion, and compensatory masquerade.[2] Examples of neuroplasticity include circuit and network changes that result from learning a new ability, information acquisition,[3] environmental influences,[4] pregnancy,[5] caloric intake,[6] practice/training,[7] and psychological stress.[8]

Neuroplasticity was once thought by neuroscientists to manifest only during childhood,[9][10] but research in the latter half of the 20th century showed that many aspects of the brain can be altered (or are "plastic") even through adulthood.[11] Furthermore, starting from

https://en.wikipedia.org/wiki/Neuroplasticity

Learning changes your brain (for the better)

- Your way of thinking has probably been permanently changed by learning about, e.g.:
  - Gravity, integrals…
  - First-order logic, computational complexity…
- This course will change your way of thinking about:
  - **Modeling the real world in a computer system**: database, but also your code
  - **Efficiently finding answers for a question** in a dataset
  - **Coordination, synchronization, contracts**

GenAI in our course

- It has mostly no place, because using it prevents the change in your thinking that we work for
- We will use it to examine its capacity to write SQL queries

Spoiler: not there yet!

Is Long Context All You Need? Leveraging LLM's Extended Context for NL2SQL

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ABSTRACT

Large Language Models (LLMs) have demonstrated impressive capabilities across a range of natural language processing tasks. In particular, improvements in reasoning abilities and the expansion of context windows have opened new avenues for leveraging these powerful models. NL2SQL is challenging in that the natural language question is inherently ambiguous, while the SQL generation requires a precise understanding of complex data schema and semantics. One approach to this semantic ambiguous problem is to provide more and sufficient contextual information.

In this work, we explore the performance and the latency trade-offs of the extended context window (a.k.a., long context) offered by Google's state-of-the-art LLM (*gemini-1.5-pro*). We study the impact of various contextual information, including column example values, question and SQL query pairs, user-provided hints, SQL

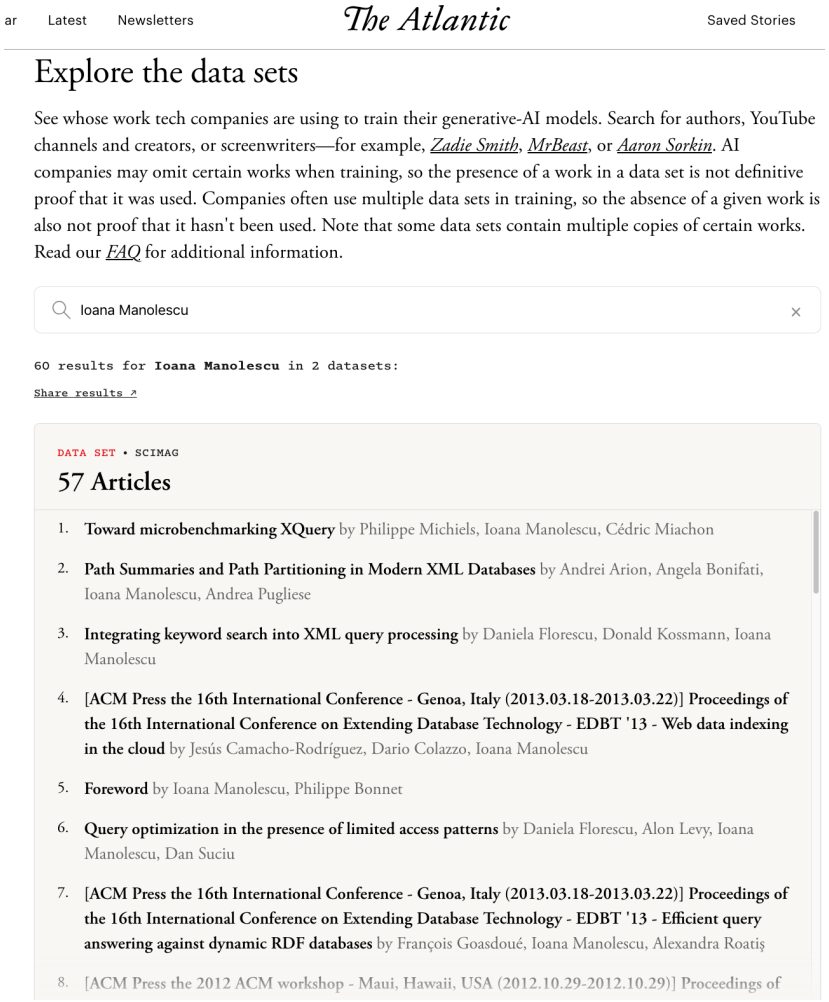
Table 1: Performance comparison of different published NL2SQL methods on BIRD dev. This excludes undisclosed methods from the leaderboard.

Method	Ex Acc (%)	Fine-tuned	Self-consistency
CHASE-SQL[a] [29]	74.46	✓	✓
XiYan-SQL	73.34	✓	✓
Long Context (ours)	**67.41**	✗	✗
Distillery[c][27]	67.21	✓	✓
E-SQL[5]	65.58	✗	✗
CHESS[34]	65.00	✓	✓
MCS-SQL[17]	63.36	✗	✓
SuperSQL[20]	58.5	✗	✓

[a] 2nd on the leaderboard as of December 17th, 2024. Long context NL2SQL pipeline was used as one of the NL2SQL pipelines to generate multiple output candidates
[c] 6th on the leaderboard as of December 17th, 2024

GenAI in technical education

- Large model such as OpenAI's have been trained on **very large technical corpora**, including some of my work



ar Latest Newsletters *The Atlantic* Saved Stories

Explore the data sets

See whose work tech companies are using to train their generative-AI models. Search for authors, YouTube channels and creators, or screenwriters—for example, *Zadie Smith*, *MrBeast*, or *Aaron Sorkin*. AI companies may omit certain works when training, so the presence of a work in a data set is not definitive proof that it was used. Companies often use multiple data sets in training, so the absence of a given work is also not proof that it hasn't been used. Note that some data sets contain multiple copies of certain works. Read our *FAQ* for additional information.

Ioana Manolescu ×

60 results for **Ioana Manolescu** in 2 datasets:

Share results ↗

DATA SET • SCIMAG

57 Articles

1. **Toward microbenchmarking XQuery** by Philippe Michiels, Ioana Manolescu, Cédric Miachon
2. **Path Summaries and Path Partitioning in Modern XML Databases** by Andrei Arion, Angela Bonifati, Ioana Manolescu, Andrea Pugliese
3. **Integrating keyword search into XML query processing** by Daniela Florescu, Donald Kossmann, Ioana Manolescu
4. **[ACM Press the 16th International Conference - Genoa, Italy (2013.03.18-2013.03.22)] Proceedings of the 16th International Conference on Extending Database Technology - EDBT '13 - Web data indexing in the cloud** by Jesús Camacho-Rodríguez, Dario Colazzo, Ioana Manolescu
5. **Foreword** by Ioana Manolescu, Philippe Bonnet
6. **Query optimization in the presence of limited access patterns** by Daniela Florescu, Alon Levy, Ioana Manolescu, Dan Suciu
7. **[ACM Press the 16th International Conference - Genoa, Italy (2013.03.18-2013.03.22)] Proceedings of the 16th International Conference on Extending Database Technology - EDBT '13 - Efficient query answering against dynamic RDF databases** by François Goasdoué, Ioana Manolescu, Alexandra Roatiş
8. [ACM Press the 2012 ACM workshop - Maui, Hawaii, USA (2012.10.29-2012.10.29)] Proceedings of

… and many technical textbooks!

→ Good LLM performance on textbook material = testing on training data (capital sin in AI ☹)

DATA SET • LIBRARY GENESIS

3 Books

1. **Web Engineering: 21st International Conference, ICWE 2021, Biarritz, France, May 18–21, 2021, Proceedings (Lecture Notes in Computer Science, 12706)** by Marco Brambilla, Richard Chbeir, Flavius Frasincar, Ioana Manolescu
2. **Web Data Management and Distribution** by Serge Abiteboul, Ioana Manolescu, Philippe Rigaux, Marie-Christine Rousset, Pierre Senellart
3. **Web Data Management** by Serge Abiteboul, Ioana Manolescu, Philippe Rigaux, Marie-Christine Rousset, Pierre Senellart

https://www.theatlantic.com/category/ai-watchdog

GenAI use in our programs

Ecole Polytechnique “Règlement des études promotion 2023”, modificatif no. 2, juin 2025

Every PC is an opportunity to evaluate you → GenAI use is by default **not allowed** during PC. (Also: prevent changes in reasoning, etc.).

Avertissement général – Plagiat

L'attention des élèves est attirée sur la gravité de l'utilisation du travail d'autrui, quelles qu'en soient les modalités (copie, paraphrase, citation, traduction…), l'origine (papier, web, communication orale…) et la nature (texte, graphique, audio, vidéo…) sans indication des sources. Cela concerne également l'utilisation de modèles d'intelligence artificielle type Large Language Model (ex de LLM : ChatGPT). L'originalité de tout travail remis par un élève pourra faire l'objet de vérifications par le corps enseignant et un logiciel spécialisé

5.2 L'utilisation de l'Intelligence Artificielle

Dans le cadre d'une évaluation, quelle que soit sa forme (évaluation sur table, devoir à la maison, projet, rapport de stage, etc.) l'utilisation d'une intelligence artificielle (IA) générative est interdite sauf si l'enseignant l'a explicitement autorisée dans ses consignes écrites. Tout manquement sera considéré comme une fraude.

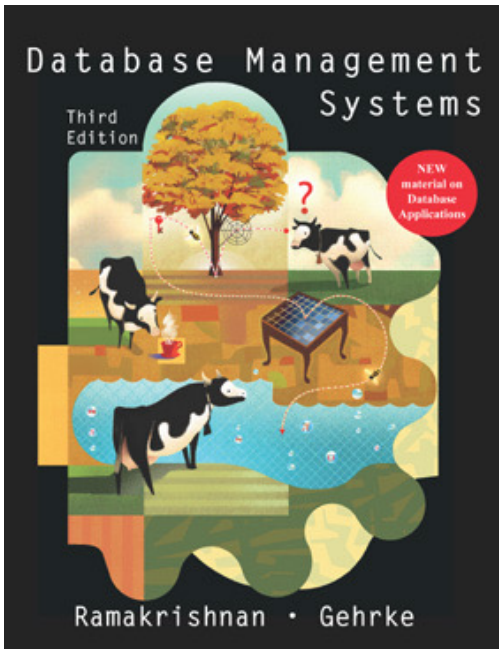
Dans le cas où l'IA générative est explicitement autorisée, l'étudiant qui en fait usage devra impérativement mentionner son utilisation avec les règles standards de citation.

en cause le tra-
ı de sources de
ıprunt doit être
ıant son origine
ilisation. En cas
ıvités à prendre
ignants.

Un constat de plagiat donne lieu à l'attribution de la note F (cf. section suivante) qui implique un passage en jury. La commission d'éthique se réunira en outre pour exa-

Reglèment des études, Master DataAI, IP Paris

Textbooks



Main book:

Database Management Systems

3rd Edition

Raghu Ramakrishnan and J. Gehrke

Other useful books:

- Database Systems: The Complete Book, Garcia-Molina, Ullman, Widom
- Database System Concepts, Korth, Sudarshan and Silberschatz
- Foundations of Databases, Abiteboul, Hull and Vianu

DATABASES AND DATABASE MANAGEMENT SYSTEMS

Databases and DBMSs

- A **database** is a large, integrated collection of data
- A **database management system** (DBMS) is a **software system** designed to store and manage large amounts of structured data
  - Declarative interface to define data stored, add data, update data, and query data
  - **Efficient querying**
  - Concurrent users
  - Reliable storage and crash recovery
  - Access control…

Database Management Systems

- Fundamentals CSC_51053
  - Data modeling
  - Relational design
  - SQL query language
- Database implementation
  - Storage, indexing
  - Query processing
  - Query optimization
  - Transaction management
- Many advanced topics
  - *Non-relational (tree- or graph- structured) data*
  - *Distributed databases, column stores*
  - *Cloud-based data management*
  - *Knowledge (not just data) bases*

Short "Beyond" lecture segments

Early DBMSs

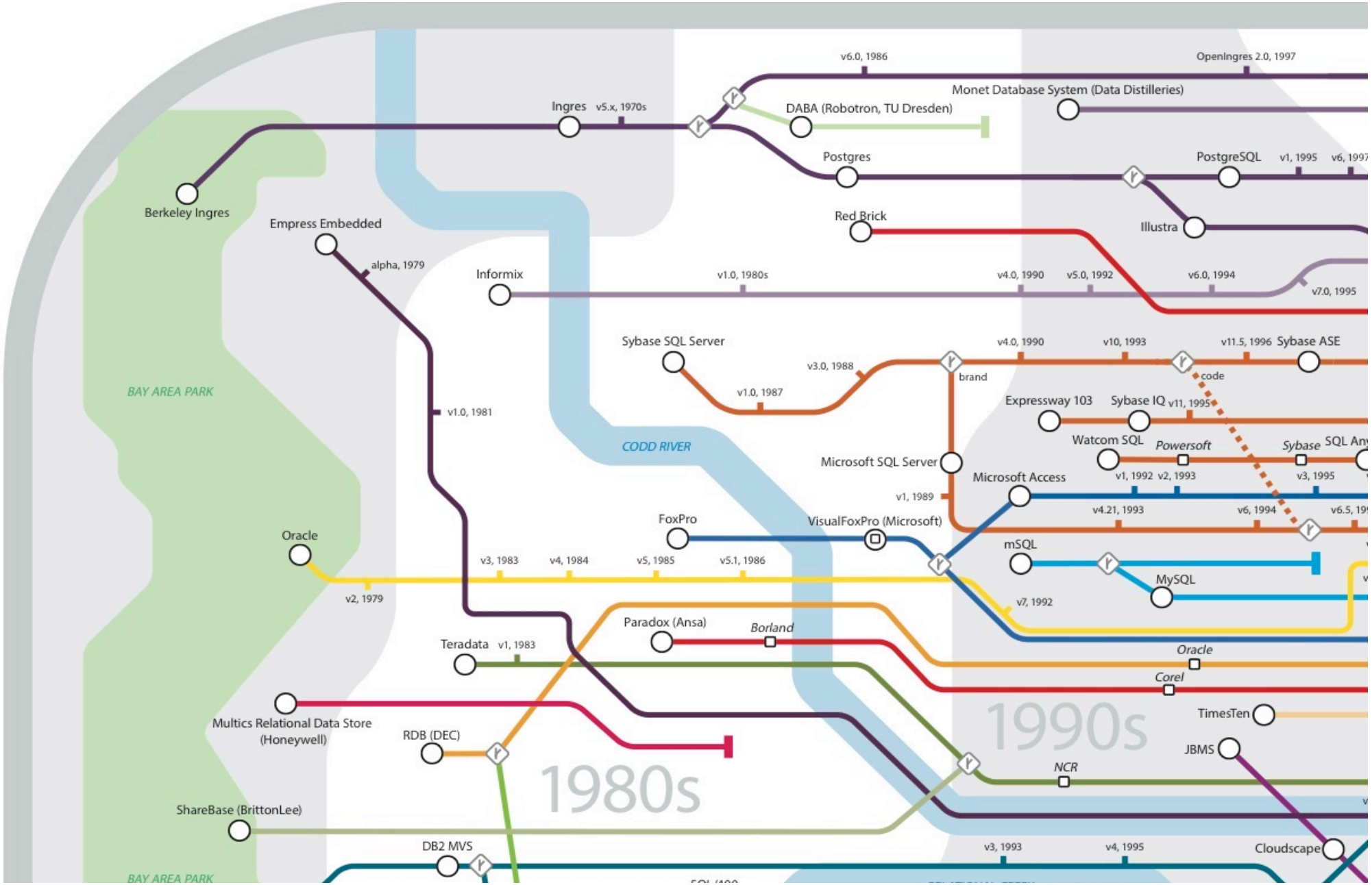
- Early DBMSs (1960's) evolved from file systems
- Many small data items, many queries and updates
  - Banking
  - Airline reservations
- 1960's "navigational DBMS"
  - Tree-based or graph-based data model
    - Like objects in a programming language, but persistent
  - Manual navigation to find what you want
  - No support for search ("write your own program")

Relational DBMS (RDBMS)

- Relational model (E.F. Codd, 1970)
  - Data independence: hides details of physical storage from users
  - Declarative query language: say **what** you want, not **how** to compute it
  - Mathematical foundation: what queries mean, possible implementations
- Query optimization (1970's till now)
  - Earliest: System R at IBM, INGRES at UC Berkeley
  - Many techniques implemented in recent systems
  - Queries can be efficiently executed despite data independence and declarative queries!

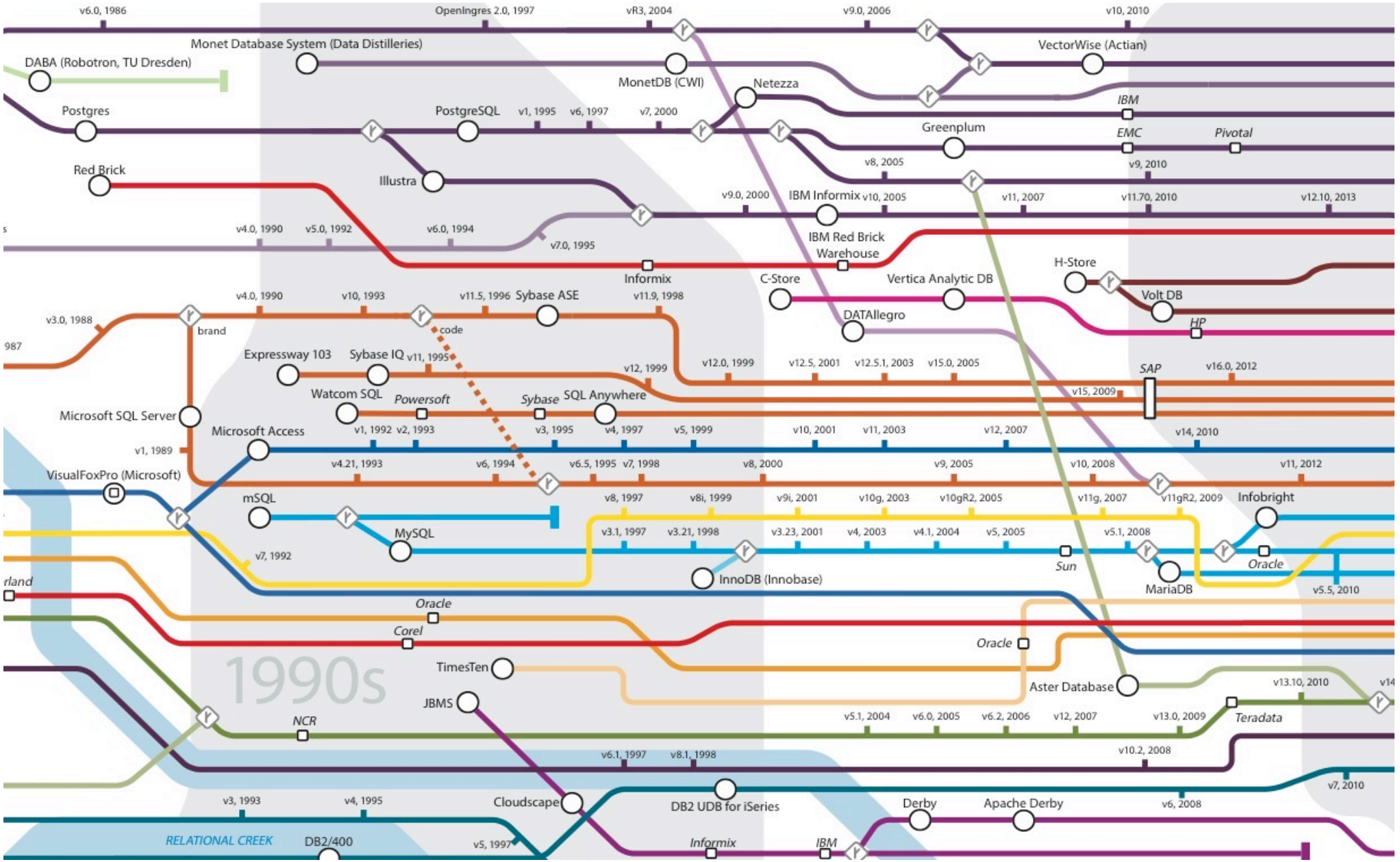
Genealogy of RDBMS systems

(c) Felix Naumann, MPI Germany



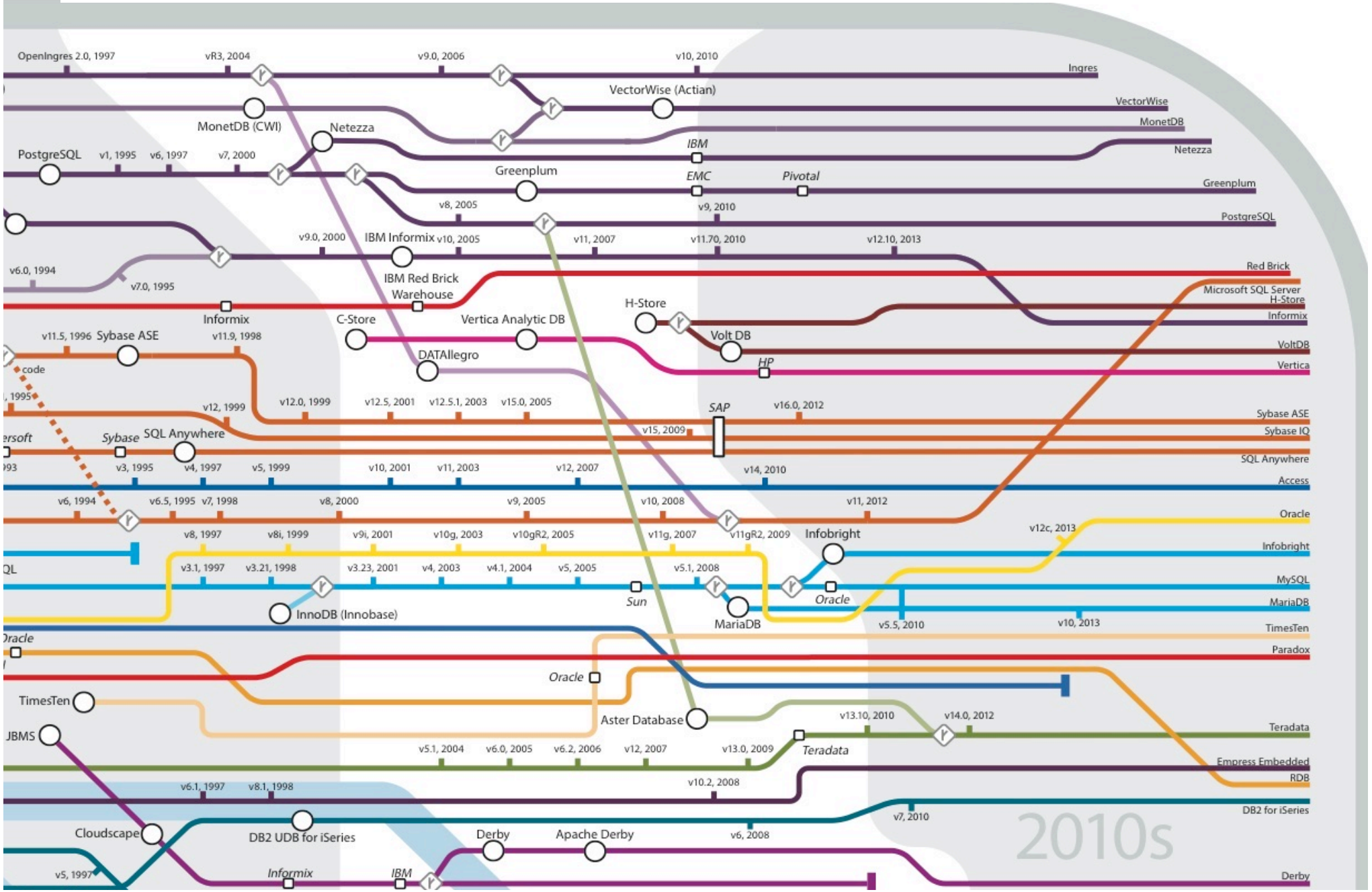
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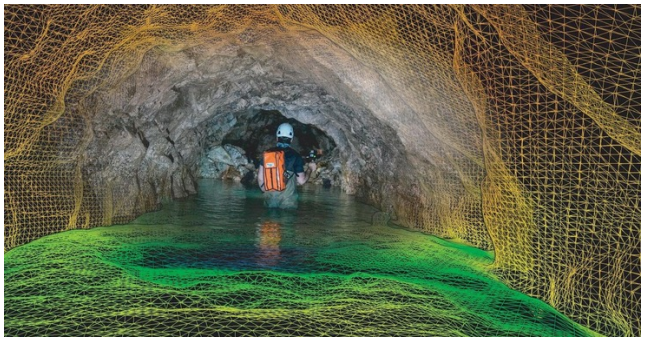
Genealogy of RDBMS systems

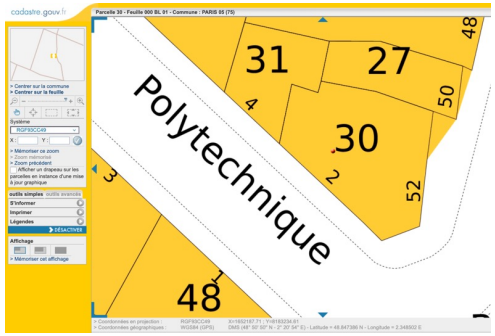
(c) Felix Naumann, MPI Germany



Typical database applications

- **Record keeping**: tracing human activity for further reference
  - Business management: **inventory of goods**, **capital flows**
  - Any public or private **accounting**
  - **Sales**, e.g., Walmart, Carrefour, Amazon, etc.
  - Typical questions: how many… are there / have been bought/sold/spent/paid… ; who bought/sold/…
- **Describing** the **physical world**
  - Geographic databases, e.g., https://cadastre.gouv.fr
  - Science observation databases, e.g., https://www.sdss.org/
  - Soil, underground, and subterranean water: https://brgm.fr



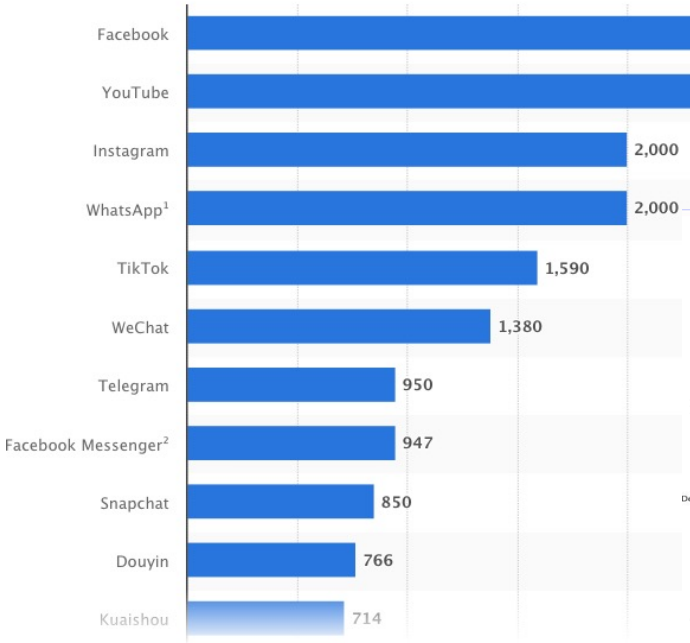


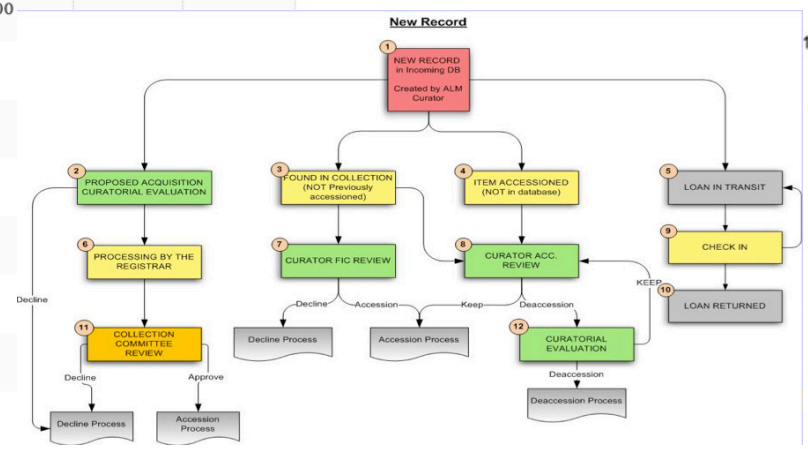


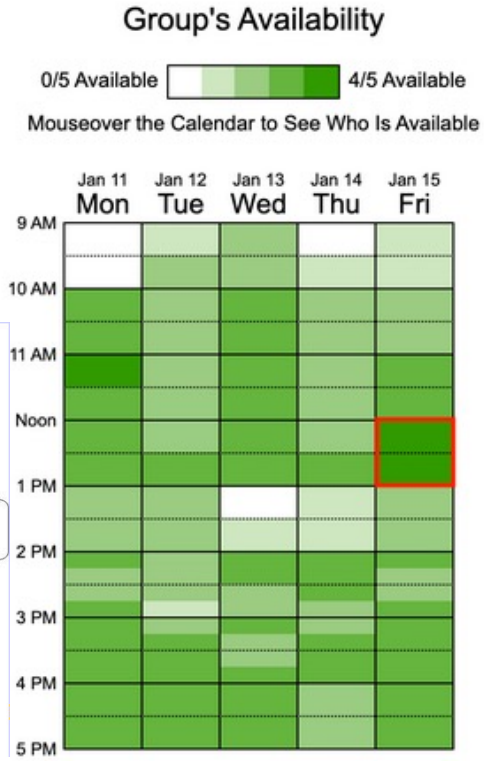
Human activity crucially enabled by databases

- **Online commerce** of goods and services
- **Social networks**
- **Multi-user processes**
  - **Online applications** (for a degree, a diploma, etc.: SCEI, Synapses…)
  - **Organization processes:** contributing information and/or making choices together, e.g., internships



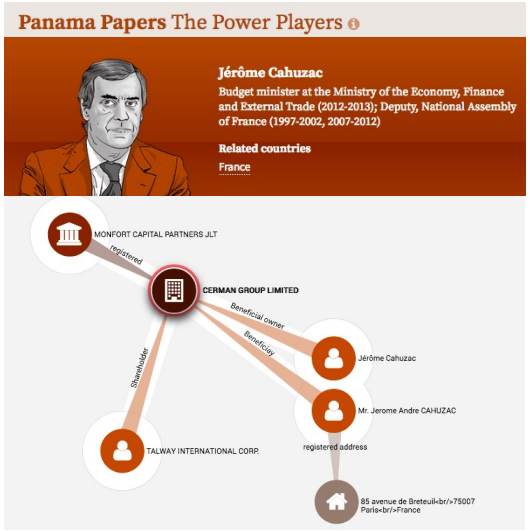


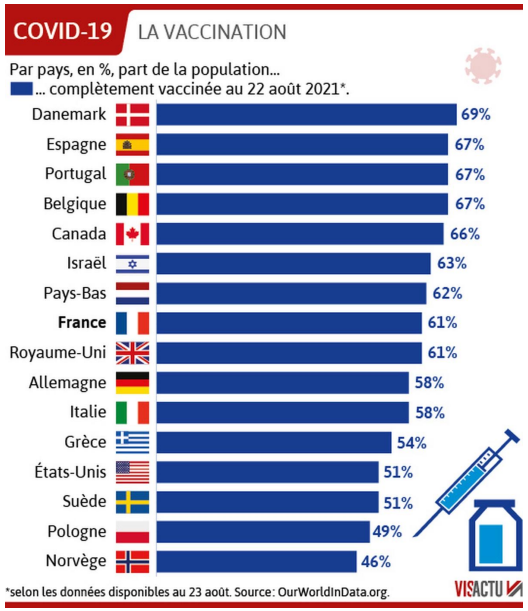




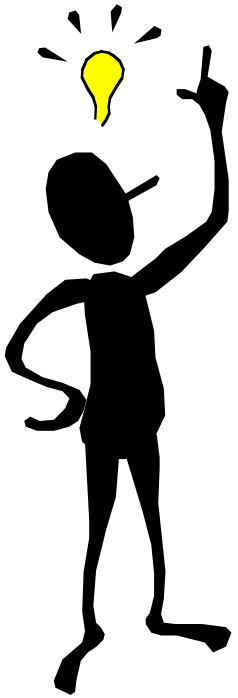
Recently Growing Database Applications

- **Graph databases**
  - Social networks, transport networks, gene-protein interaction, data journalism...
  - *Querying, aggregation, integration, ranked search*
- **Health**
  - Personal medical records
  - Evolution along time
  - Statistics over populations
- **Knowledge bases**
  - Data + knowledge
  - Heterogeneous data
  - *Extraction from text, reconciliations...*





How does one build a database?



Example: The Internet Shop

- DBDudes Inc: a well-known database consulting firm
- Barns and Nobble (B&N): a large bookstore

- B&N decides to go online but needs help!

- Step 1: DBDudes and B&N sit down for a requirement analysis

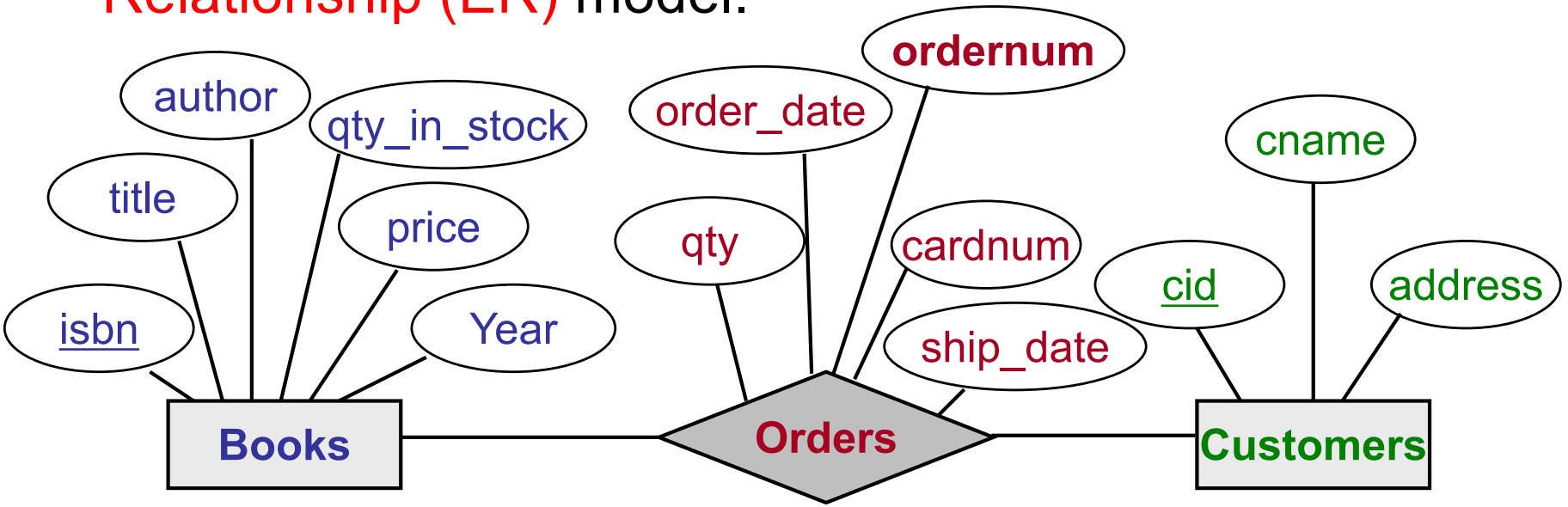
Step 1: Requirements Analysis

"I'd like my customers to be able to browse my catalog of books and place orders online."

- **Books**:
  - For each book, B&N's catalog contains its ISBN number, title, author, price, year of publication, …
- **Customers**:
  - Most customers are regulars with names and addresses registered with B&N.
  - New customers must first register online.
- **On the new website**:
  - Customers identify themselves before browsing and ordering.
  - Each order contains the ISBN of a book and a quantity.
- **Shipping**:
  - For each order, B&N ships all copies of a book together once they become available.

Step 2: Conceptual Design

- A high level description of the data in terms of the Entity-Relationship (ER) model.



- Design review:
  - What if a customer places two orders of the same book in one day?
  - What if a customer wants to have different books in the same order?

Step 3: Logical Design

- Mapping the ER diagram to the relational model

```
CREATE TABLE Books
(isbn          CHAR(10),
 title         CHAR(80),
 author        CHAR(80),
 qty_in_stock  INTEGER,
 price         REAL,
 year          INTEGER,
 PRIMARY KEY(isbn))
```

```
CREATE TABLE Customers
(cid           INTEGER,
 cname         CHAR(80),
 address       CHAR(200),
 PRIMARY KEY(cid))
```

```
CREATE TABLE Orders
(ordernum INTEGER,
 isbn          CHAR(10),
 cid           INTEGER,
 cardnum CHAR(16),
 qty           INTEGER,
 order_date DATE,
 ship_date     DATE,
 PRIMARY KEY(ordernum, isbn),
 FOREIGN KEY (isbn)
     REFERENCES Bo
 FOREIGN KEY (cid)
     REFERENCES Cu
```

```
CREATE VIEW OrderInfo
(isbn, cid, qty, order_date, ship_date)
AS SELECT O.isbn, O.cid, O.qty,
                  O.order_date, O.ship_date
      FROM    Orders O
```

- Access control: use views to restrict the access of certain employees to customer sensitive information

Step 4: Schema Refinement

Orders

ordernum	isbn	cid	cardnum	qty	order_date	ship_date
120	0-07-11	123	40241160	2	Jan 3, 2006	Jan 6, 2006
120	1-12-23	123	40241160	1	Jan 3, 2006	Jan 11, 2006
120	0-07-24	123	40241160	3	Jan 3, 2006	Jan 26, 2006

Redundant Storage

Orders

ordernum	cid	cardnum	order_date
120	123	40241160	Jan 3, 2006

Orderlists

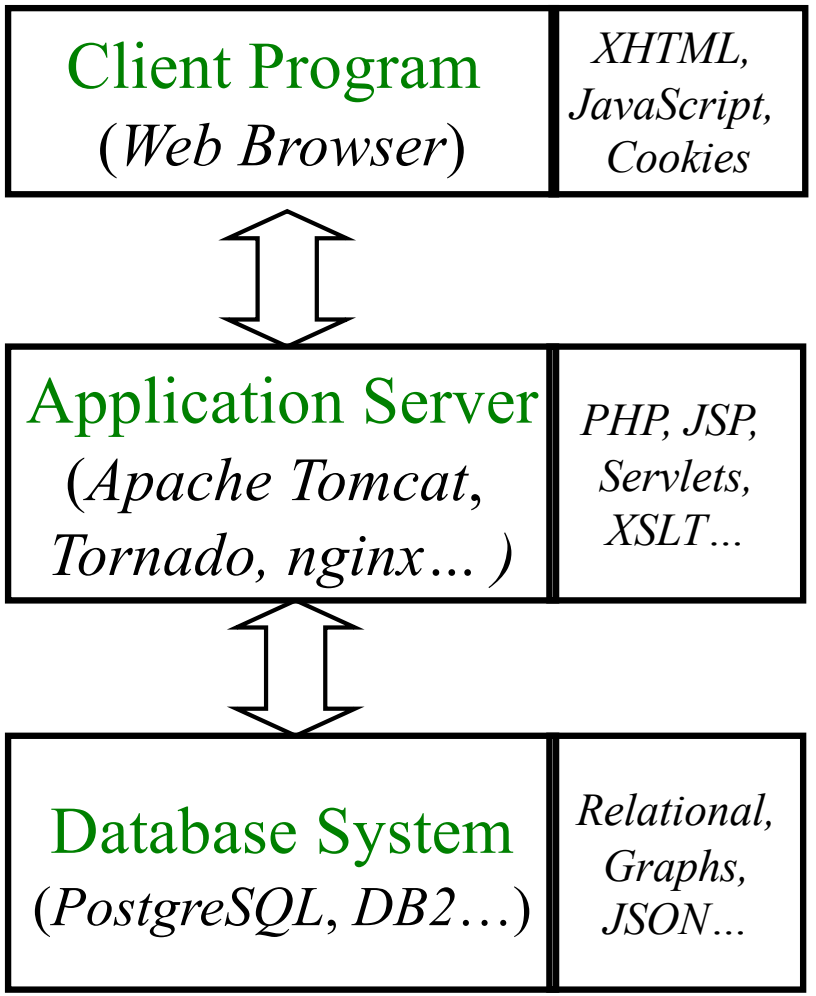
ordernum	isbn	qty	ship_date
120	0-07-11	2	Jan 6, 2006
120	1-12-23	1	Jan 11, 2006
120	0-07-24	3	Jan 26, 2006

Step 5: Internet Application Development

Presentation tier

Application logic tier

Data management tier

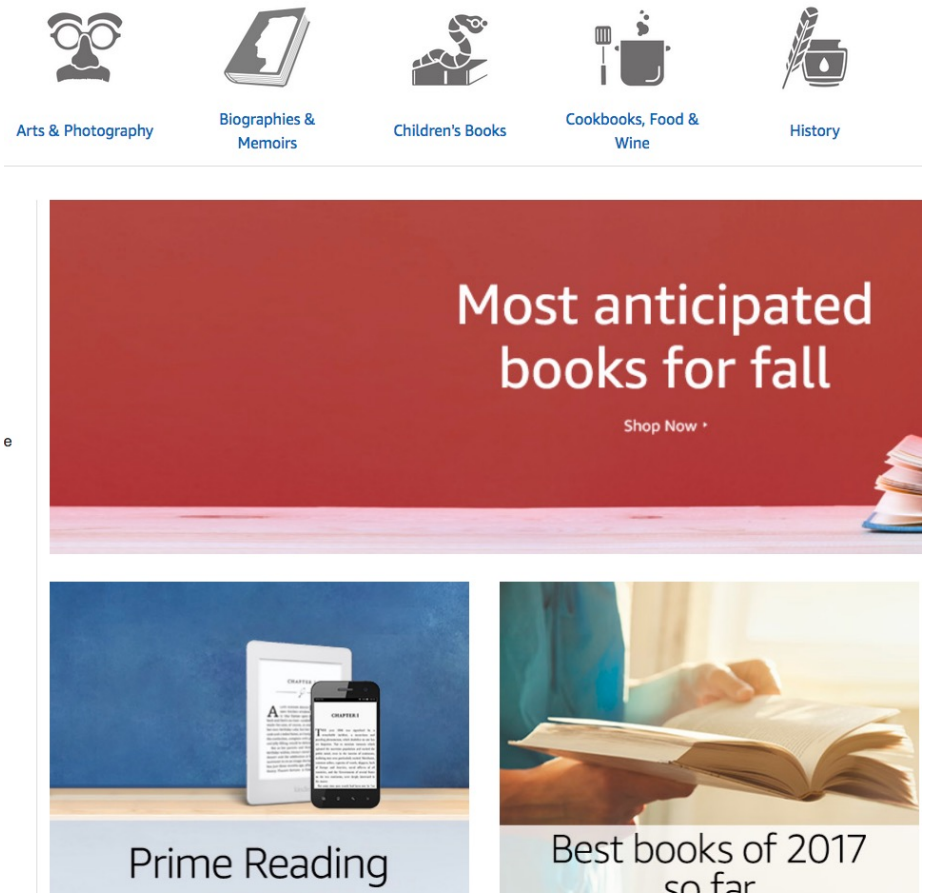


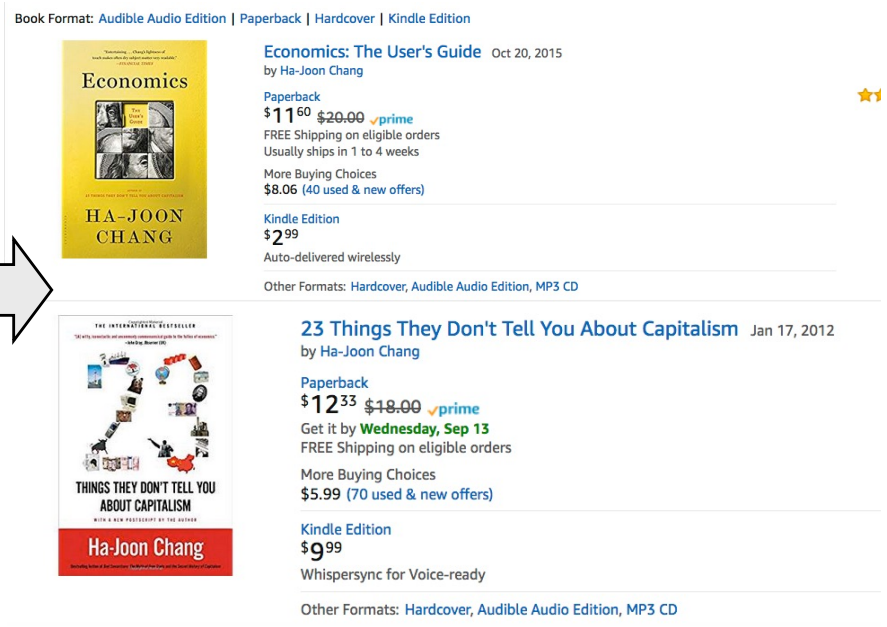
B&N Client:
- User input
- Display of output

B&N Business logic:
- **Home page**
- **Login page**
- **Search page**
- **Cart page**
- **Confirm page**

B&N Data:
- **Books**
- **Customers**
  (User login)
- **Orders**
- **Orderlists**

An Example Internet Store





Example SQL Queries

In these examples, queries have *parameters*.
These will be instantiated when the queries are called from the application.

Login Page

```
SELECT  cid, username, password
FROM    Customers
WHERE   username = '<SpecifiedUsername>'
```

Get Book Details Page

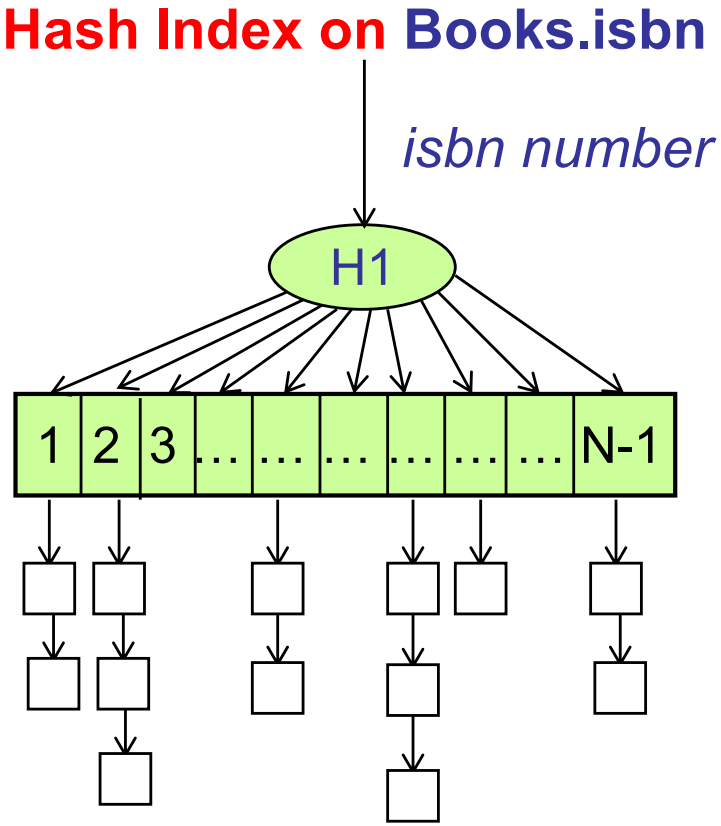
```
SELECT  isbn, title, author, price
FROM    Books
WHERE   isbn = '<GivenISBN>'
```

Step 6: Physical Design

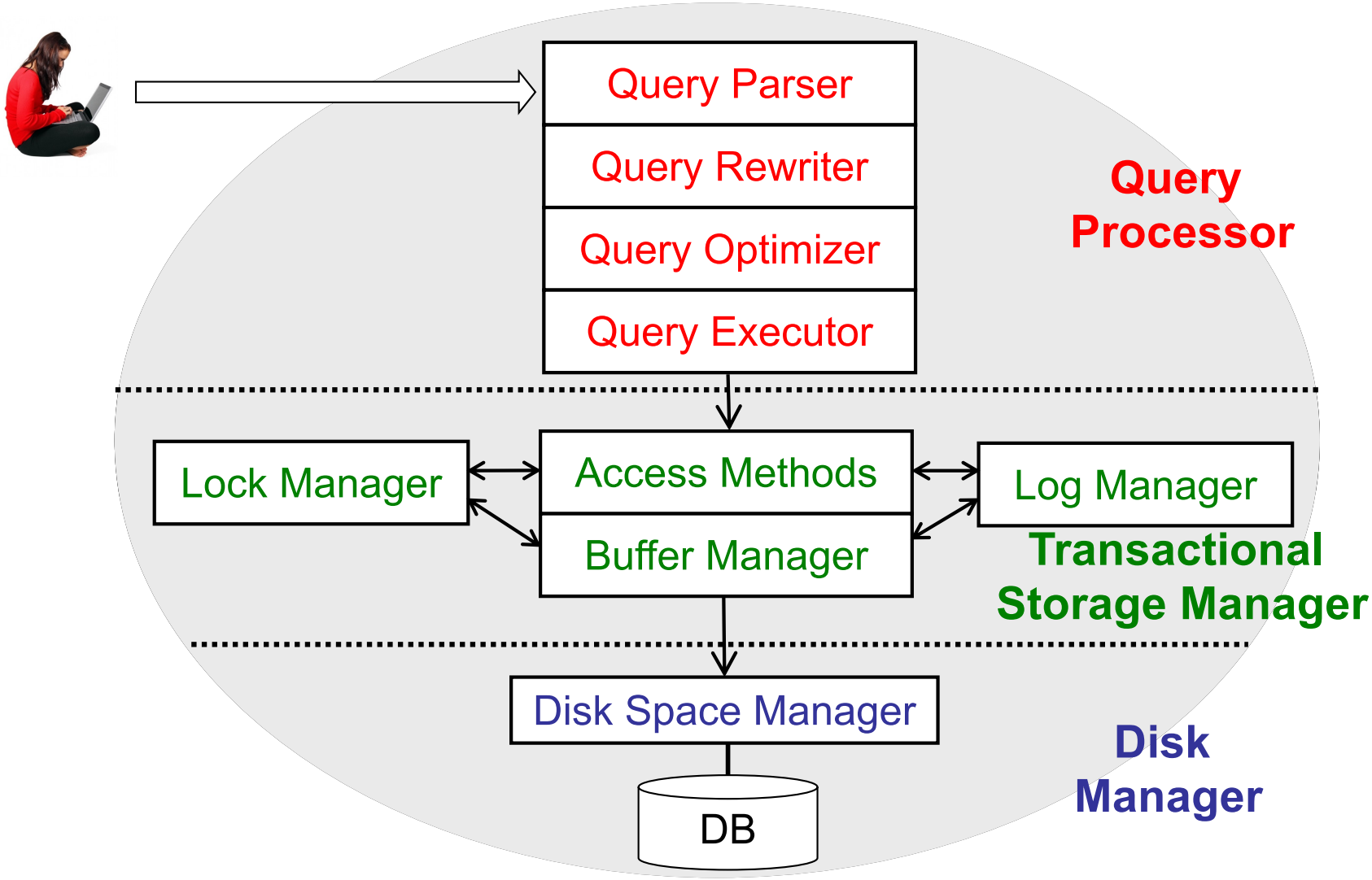
- Auxiliary data structures, indexes, to **speed up searches**, e.g., hash index, B+tree, R-tree

Books

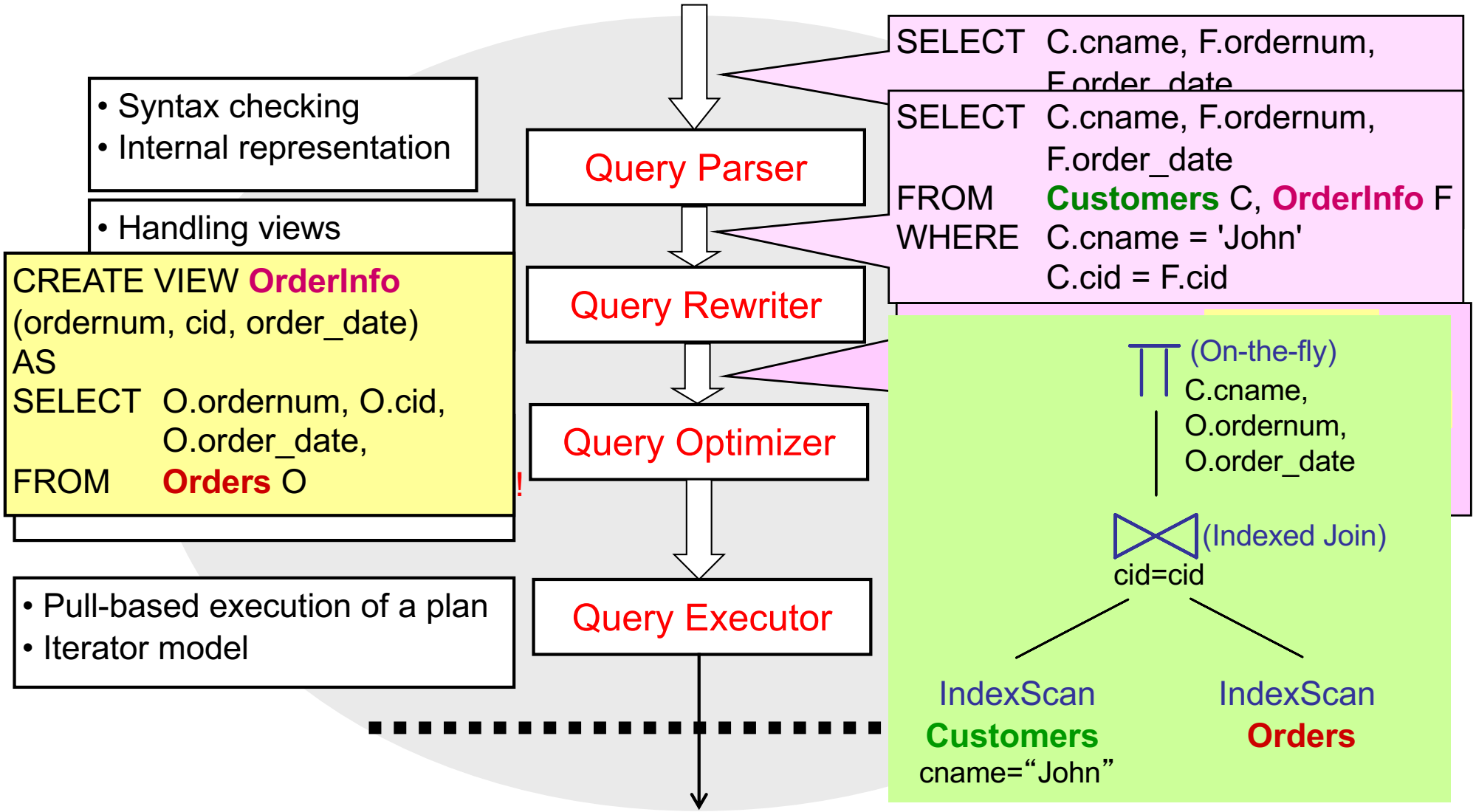
isbn	title	author	price	year	qty
0-07-11	egacies of the Turf	Edward L. Bowen	29.95	2003	10
1-12-23	attle Slew	Dan Mearns	24.95	2000	0
0-07-24	pectacular Bid	Timothy Capps	16.95	2001	3
…	…	…	…	…	…
…	…	…	…	…	…
…	…	…	…	…	…
…	…	…	…	…	…
…	…	…	…	…	…



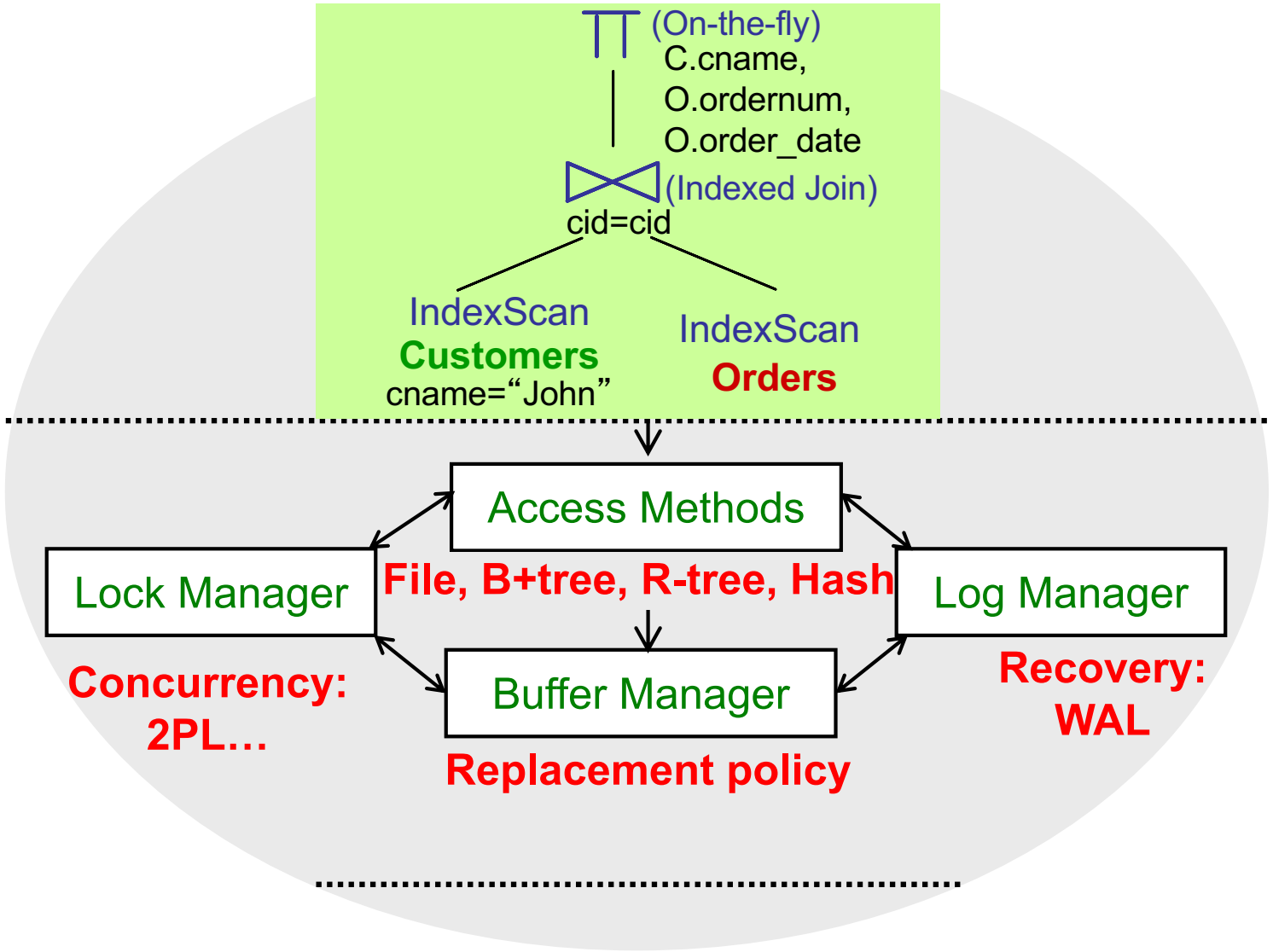
DBMS Internal Architecture



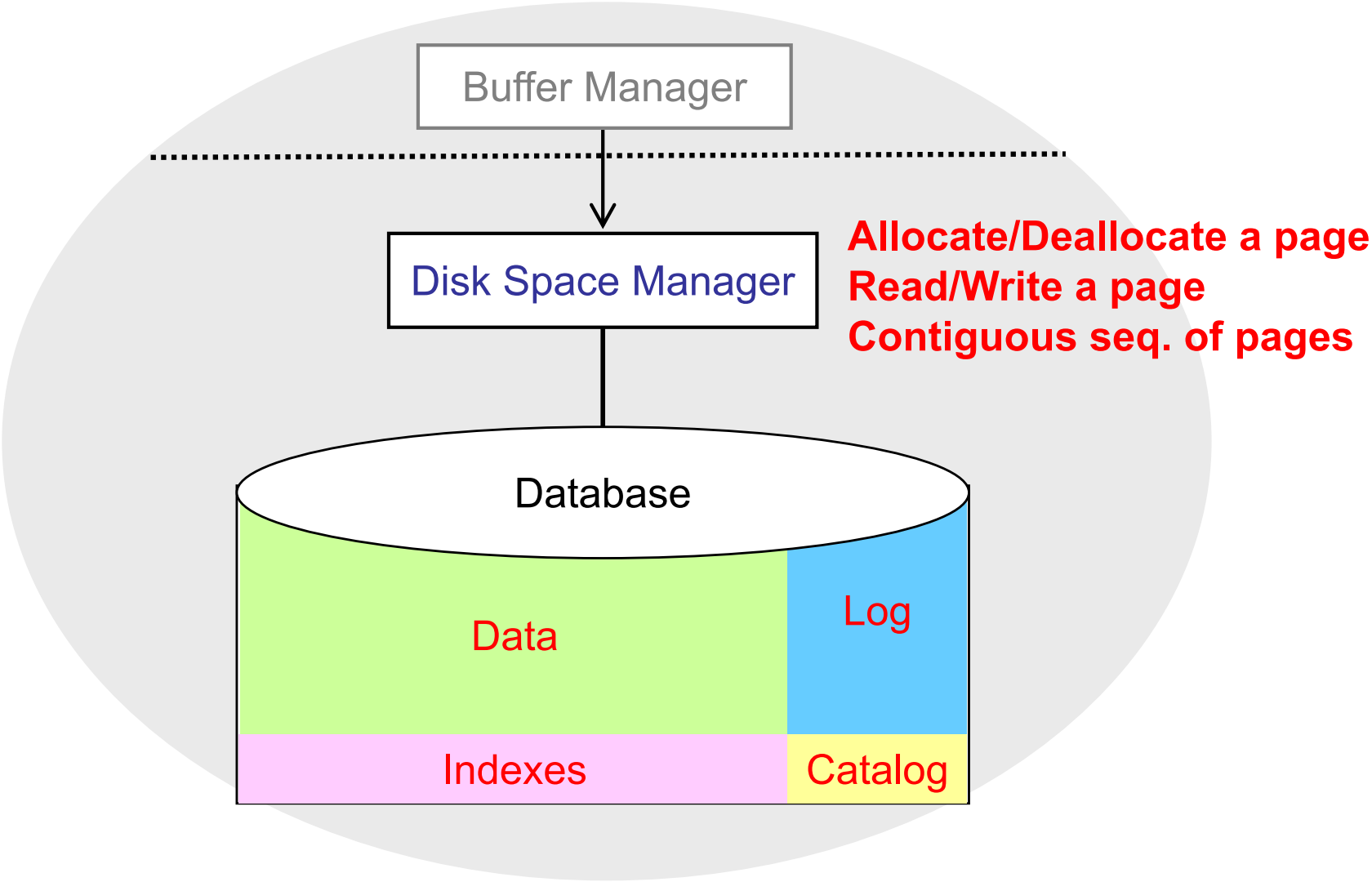
Query Processor



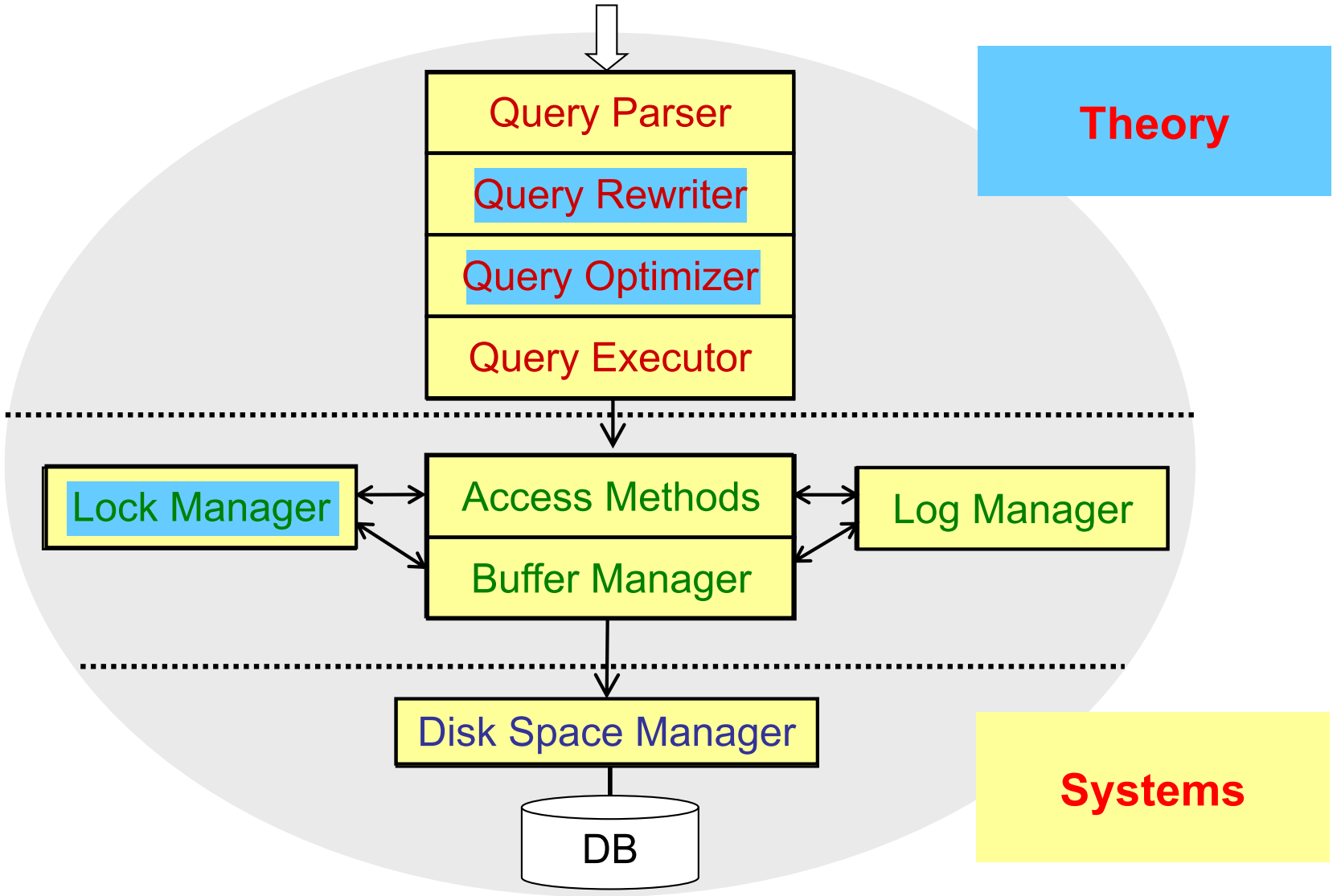
Transactional Storage Manager



Disk Manager



DBMS: Theory + Systems



Questions?

